

One blocking step: the obstruction is the variation, not the size, of the inter-block coupling

Partition the lattice into blocks of M^3 sites and let P project onto the tensor product of the low-energy subspaces of the blocks. The transfer lemma then reduces the gap of the whole system to the gap of the blocked Hamiltonian PHP minus the Schur error of the straddling plaquettes. Counting gives, per block, $6M^2$ straddling plaquettes of norm $\|w_p\| \leq 4N\hbar c/(ag^2)$ each, against an intra-block gap $\Delta_M = (\hbar c/a)\delta(g; M)$, so the naive operator-norm estimate gives a Schur error of order

$$\frac{\|B\|^2}{\Delta_M} \sim 36M^4 \frac{\|w\|^2}{\Delta_M} = 36M^4 \frac{\hbar c}{a} \frac{16N^2}{g^4 \delta(g; M)},$$

which exceeds the gap it is meant to preserve by a factor of order $M^4 N^2/(g^4 \delta^2)$. Lemma 1' of the Schur note shows where that estimate is wasteful: a part of the Schur term proportional to the identity shifts all levels equally and costs nothing, and the boundary energy is exactly of that kind. What the induction needs is the **variation** of the Schur term over the low-energy subspace, for which a connected estimate replaces $\|B\|^2 = (\sum_p \|w_p\|)^2$ by $\sum_p \|w_p\|^2$ and so replaces M^4 by M^2 . Under that replacement one blocking step closes when

$$\frac{\|w\|}{\Delta_M} \lesssim \frac{1}{M\sqrt{6}} \quad \text{i.e.} \quad g^4 \gtrsim \frac{8\sqrt{6}MN}{\gamma C_2(R_{\min})},$$

a fixed threshold at fixed block factor. Since the effective coupling grows toward the infrared, the threshold, once reached, is satisfied at every later step, so the induction closes from that scale onward. The number of doublings needed to reach it from the cutoff is

$$n \simeq \frac{1}{2b_0 \log 2} \left(\frac{1}{g_{\text{UV}}^2} - \frac{1}{g_{\text{thr}}^2} \right), \quad b_0 = \frac{11N}{48\pi^2},$$

finite and of the order of tens for realistic couplings. The whole difficulty of the mass gap is therefore concentrated in a **finite number of blocking steps in the intermediate coupling regime**, each of which generates couplings beyond the Kogut–Susskind form that must be carried along. Constants explicit; the connected estimate is stated as the required input and is not proved here; nothing promoted.

1. The block decomposition and its counting

Fix a block factor $M \geq 2$ and partition the periodic lattice of N_s^3 sites, $N_s = MK$, into K^3 blocks of M^3 sites. Assign each link to the block containing its midpoint, so the links are partitioned. A plaquette is *interior* when all four of its links lie in one block and *straddling* otherwise. Write

$$H = \sum_{\alpha} H_{\alpha} + \sum_{p \text{ straddling}} w_p, \quad H_{\alpha} = \frac{\hbar c g^2}{2a} \sum_{\ell \in \alpha} (-\Delta_{\ell}) + \sum_{p \subset \alpha} w_p.$$

Counting. Each block has six faces of M^2 plaquettes, so

$$\#\{\text{straddling plaquettes touching one block}\} = 6M^2, \quad \|w_p\| \leq \frac{4N\hbar c}{a g^2},$$

with N replaced by $\dim_{\mathfrak{f}}$ for a general compact group, as in the Lieb–Robinson note §1. Each H_{α} is a Kogut–Susskind Hamiltonian on a block with free boundary, so T1 applies to it: discrete spectrum, unique positive ground state Ω_{α} , gap $\Delta_M = (\hbar c/a)\delta(g; M)$.

Let Π_{α} project onto the spectral subspace of H_{α} below a cutoff E_c chosen in the gap region, and

$$P = \bigotimes_{\alpha} \Pi_{\alpha}, \quad A = PHP, \quad D = \bar{P}H\bar{P} \geq \sum_{\alpha} E_0^{\alpha} + E_c.$$

The compression A is the blocked Hamiltonian: a Hamiltonian on the product of block low-energy spaces, whose gap the induction must track.

2. Why the naive estimate fails

The coupling operator is $B = PH\bar{P}$, bounded by the sum of the straddling terms touching the blocks involved,

$$\|B\| \leq \sum_{p \text{ straddling}} \|w_p\| \sim 6M^2 \cdot \frac{4N\hbar c}{ag^2} \quad \text{per block},$$

and the transfer lemma of the Feshbach note charges $\|B\|^2/(E_c)$ against the gap. With $E_c \sim \Delta_M$,

$$\frac{\|B\|^2}{\Delta_M} / \Delta_M \sim 36M^4 \left(\frac{\|w\|}{\Delta_M} \right)^2 = 36M^4 \left(\frac{8N}{\gamma C_2 g^4} \right)^2,$$

using $\Delta_M \geq \gamma(g^2/2)C_2\hbar c/a$ from the T2 note at strong coupling. The ratio exceeds one unless $g^4 \gtrsim M^2N$, and at weak coupling it is enormous. Taken at face value, real-space blocking is useless.

3. What the estimate is actually measuring

The quantity $\|B\|^2/\Delta_M$ is dominated by the **boundary energy**: the second-order effect of the straddling plaquettes lowers the ground state of the coupled system by an amount proportional to the total boundary area, which is extensive in M^2 per block and has nothing to do with the gap. Lemma 1' of the Schur note isolates that contribution: if

$$S(E) = B(D - E)^{-1}B^* = \eta_0(E)\mathbb{1} + R(E), \quad 0 \leq R(E) \leq \epsilon(A - a_0) + \eta_1,$$

then the constant η_0 drops out of the gap entirely and only ϵ , η_1 and the slope η'_0 enter. The boundary energy is the constant; what survives is the dependence of the boundary energy on **which** low-energy state of the blocks the system occupies.

The required input. For each straddling plaquette separately, $w_p(D - E)^{-1}w_p$ has norm at most $\|w_p\|^2/E_c$, and its variation across $\text{ran } P$ is at most twice that. Cross terms between two straddling plaquettes $p \neq p'$ contribute to the constant when they are far apart and to the variation only through connected configurations. A connected estimate of the form

$$\|R(E)\| \leq C \sum_{p \text{ straddling}} \frac{\|w_p\|^2}{E_c} \quad \text{in place of} \quad \frac{(\sum_p \|w_p\|)^2}{E_c}$$

therefore replaces M^4 by M^2 . This is exactly what a cluster expansion supplies at strong coupling, and it is the one analytic input the blocking step needs. It is stated here as a hypothesis.

4. The one-step inequality under that input

Superseded. Sections 4 and 5 are corrected in the blocking-criterion note: the connected estimate is available from Yarotsky's polymer expansion, and the resulting threshold is worse than the direct one by a factor $3M^2/8$, so the induction below does not close. The counting of Section 5 survives as a count of renormalization-group steps.

Proposition (conditional). Assume the connected estimate of Section 3 with constant C , and take $E_c = \Delta_M$. Then the transfer lemma gives

$$\text{gap}(H) \geq \text{gap}(A)(1 - \epsilon) - \eta_1, \quad \epsilon, \frac{\eta_1}{\Delta_M} \leq 6CM^2 \left(\frac{\|w\|}{\Delta_M} \right)^2,$$

so one blocking step preserves a positive gap whenever

$$\frac{\|w\|}{\Delta_M} \leq \frac{1}{M\sqrt{6C}}, \quad \text{that is} \quad g^4 \geq \frac{8\sqrt{6C}MN}{\gamma C_2(R_{\min})}.$$

The threshold depends on the block factor M and on the group, and on no other scale: it is a **fixed** coupling threshold once M is fixed, say $M = 2$.

5. The induction, and where the finite difficulty sits

Under blocking, the effective coupling of an asymptotically free theory grows toward the infrared. If the blocked Hamiltonian A were again of Kogut–Susskind form with a coupling $g_{n+1} > g_n$, then the threshold of Section 4, once satisfied at some step, would be satisfied at every later step, and the induction would give a gap at all larger scales, uniform in the volume. Two obligations remain.

Reaching the threshold. At the cutoff the coupling is small. With the one-loop running written for a doubling of scale,

$$\frac{1}{g_{n+1}^2} = \frac{1}{g_n^2} - 2b_0 \log 2, \quad b_0 = \frac{11N}{48\pi^2},$$

the number of doublings from g_{UV} to g_{thr} is

$$n \simeq \frac{1}{2b_0 \log 2} \left(\frac{1}{g_{UV}^2} - \frac{1}{g_{thr}^2} \right),$$

finite and logarithmic in the ratio of scales. For $SU(2)$, $2b_0 \log 2 = \frac{11}{12\pi^2} \log 2 \approx 0.064$, so a bare coupling $g_{UV}^2 = 1/2$ takes of the order of thirty doublings, a scale ratio of about 10^9 , to reach a threshold at $g_{thr}^2 \approx 20$. The mass gap therefore requires control of a **finite, explicitly bounded number of blocking steps**, all of them in the intermediate regime where neither the perturbative expansion of the Feshbach note nor the cluster estimate of Section 3 applies.

Staying in the family. The blocked Hamiltonian A is a Hamiltonian on the product of block low-energy spaces, and it contains, besides a nearest-block plaquette term, every operator generated by the compression: longer-range terms, terms of higher order in the block observables, and terms without a Kogut–Susskind counterpart. Carrying a controlled family of such Hamiltonians through the steps is the constructive renormalization-group problem that Balaban’s programme solves on the Euclidean side in finite volume with an ultraviolet cutoff, and which remains open beyond that.

6. What this step establishes

The blocking route survives the first objection: the enormous inter-block coupling enters the gap only through its variation, and the boundary energy, which is what makes $\|B\|$ large, is free by Lemma 1’. The route then needs one analytic input, a connected estimate for the Schur variation, and one structural input, a stable family of effective Hamiltonians. The first is a cluster expansion and is available at strong coupling; the second is the open problem. The quantitative shape of the difficulty is now explicit: a fixed coupling threshold $g_{thr}^4 \simeq 8\sqrt{6}CMN/(\gamma C_2)$, and a finite number of steps, logarithmic in the scale ratio, to reach it.

7. Consequence for STATE

Step 2 of the foreseen route is written out with its constants. The two inputs it needs are named and separated: a connected (cluster) estimate for the variation of the Schur term, which would make the one-step inequality unconditional at strong coupling, and a stable family of blocked Hamiltonians, which is the constructive renormalization-group problem. The next tractable item is the first of these at strong coupling, where Yarotsky’s expansion already produces connected estimates of the required kind and may be readable as such directly.